The Next Generation Sequencing Quality Initiative

The Next Generation Sequencing (NGS) Quality Initiative is a collaboration between the Centers for Disease Control and Prevention (CDC), the Association of Public Health Laboratories (APHL), and state and local public health laboratories (PHLs) to address the many challenges laboratories encounter when implementing NGS-based assays. The Initiative is developing an NGS-focused quality management system (QMS) to assure foundational quality during the development and implementation of sequencing-based tests by providing customizable, ready-to-implement tools and resources that laboratories can use to standardize and institute quality management practices and procedures. The NGS Quality Initiative has published additional tools and resources, including templates and procedures, that may be of assistance to laboratories throughout their NGS workflow. Please visit the following website to access these resources: https://www.cdc.gov/labquality/qms-tools-and-resources.html.

This document is intended to be used as a tool for implementing, improving, or maintaining an NGS QMS. Blue text provides examples for appropriate input and can be changed, deleted, or augmented as needed for the laboratory's specific requirements.

These documents and tools are not controlled files; format and content **must** be modified as needed to meet the document control, QMS, or regulatory requirements within your laboratory. It is the responsibility of your laboratory to take any necessary actions to ensure the information within these documents remains applicable.

Insert Laboratory-Specific Name Here

NGS Method Validation Plan

Template

1.0 **Purpose of Validation**

0.1 Type of Validation:

Initial validation (new test procedure)

Technical Modification

Modification to procedure

Relocation of equipment

Change to instrument hardware or software

Change in bioinformatics pipelines (i.e., software, scripts, reference databases)

Change in reagent formulation

Change in reagent manufacturer

Change in patient population

Change in intended use

Other___________________________________

0.2 Statistical methods defined in this plan are designed to address the specific needs of the validation. The (comparison or reference method or material) will be used for evaluating performance. (Method name) is expected to (describe improvement or how the new method differs from established testing).

2.0 **Scope**

1.0

1.1 **Method Validation Plan for:** *[insert method name]*

1.2 **Branch/Laboratory:** *[branch and laboratory name]*

1.3 **Test System Name:** *[insert test system name, and indicate whether new or existing test system]*

1.4 **Optional: Test Order Code (Client services manual):** (Test code)

1.5 **Test Procedure Document Number(s) and Revision Number:**

Document Name	Document Number	Revision Number	Effective Date
[insert laboratory-specific document name here]	*[insert laboratory-specific document n umber here]*	*[insert laboratory-specific revision number here]*	*[insert laboratory-specific effective date here]*

3.0 **Summary of the Test Procedure Purpose/Principle:**

2.0

2.1 **Assay result type** (choose one):

Qualitative

Qualitative-titered

2.2 **Assay regulatory type, as applicable** (choose one)**:**

Laboratory developed test (LDT)

Modified FDA-cleared/approved

Not regulated (e.g., surveillance)

2.3 **Agent or analyte detected by the method:** (insert name)

2.4 **How test results are to be used:** Describe how the test results will be used. Include the following elements as applicable: (1) Presumptive, screening, monitoring, confirmatory (e.g., presumptive to detect infection, screening test to rule out disease (high sensitivity, low specificity), a confirmatory test (high specificity), to monitor treatment response, to characterize or phenotype a pathogen, a research trial or surveillance activity falling under CLIA); (2) Detail if the results are used alone, or in conjunction with other assays as part of a specific testing algorithm and the extent to which interpretation needs to be in conjunction with clinical signs and symptoms (e.g., Stand-alone test, used in conjunction with other assays (list related documents title and number); (3) Specify if result use differs among sample types or among patient populations; (4) If the test is used solely for research or surveillance purposes (not CLIA), specify that: "These results are intended for public health purposes only and shall NOT be communicated to the patient, their care provider, or placed in the patient's medical record. These results should NOT be used for diagnosis, treatment, or assessment of patient health or management."

4.0 **Specimen Types to be Validated:**

Required Information	Laboratory Specific Information
Acceptable specimen types	(List all specimen types)
Specimen matrices to be used in validation	(List all matrices to be used in the method validation, including transport media)

5.0 **Limiting factors and justification for limited sample size:**

3.0

4.0

4.1 Sample scarcity, urgent public health response, significant aliquoting/ replicate testing.

6.0 **Acceptance Criteria:**

Performance Characteristic	Comparator Method Specifications (if available)	New Test Procedure: Minimum acceptable values
Sensitivity	Enter the percentage	Enter the percentage
Specificity	Enter the percentage	Enter the percentage
Accuracy	Enter the percentage	Enter the percentage
Precision/Reproducibility	Enter the percentage	Enter the percentage
Applicable Genome Region	Enter comparator method specifications here	Enter the region of the genome in which sequence of an acceptable quality is expected to be derived
Clinical Validity (as applicable)	Positive Predictive Value (PPV) and Negative Predictive Value (NPV) of comparator method	Enter the prevalence in the relevant population(s) Enter the minimum acceptable PPV and NPV
Any other performance characteristics required for this validation (e.g., cross-reactivities, interfering substances, lower limit of detection)	(Row may be deleted if not applicable)	(Row may be deleted if not applicable)

7.0 **Reference/Normal value:**

5.0

6.0

6.1 Describe whether the analyte is expected to be present or absent in the target population(s) and any caveats (e.g., vaccine status).

6.2 Describe whether the validation study will establish a reference value (describe samples here) or describe the literature on prevalence and test performance in reference population(s).

8.0 **Sample Requirements:**

NOTE: Ideally, use specimens that have the specimen stability data in place. See Appendix B for Specimen stability.

7.0

7.1 **Sample Selection Summary:**

a. True Positive samples: Justify True Positive sample selection according to the following three criteria: Diversity, expected sample testing volume, and public health impact.

i. Spike-in samples must include those near the lower limit of detection or other clinical decision point(s).

ii. Mock clinical samples must include those with low, mid, and high ranges of analyte present, as applicable.

b. True negative samples: Justify True Negative Sample selection according to the following three criteria: agent or analyte similarity, symptom similarity, and healthy population specimens.

NOTE: If revalidating after a change to data analysis methods only (“dry lab”), the samples selected may be electronic data.

7.2 If retrospective data will be used for the validation, include description of retrospective data that will be abstracted for the following: sample selection criteria, dates of testing, and availability and location of original raw data. It must be clear in each section below whether data are retrospective or prospective.

NOTE: The CLIA Lab Director may require consultation for the use of retrospective data.

7.3 **Clinical Samples:** Limiting factors and justification for limited true positive sample size: Sample scarcity, urgent public health response.

Clinical Samples	Description
Total positive samples	#
Total negative samples	#
Sample volume (units)	#
Sample source/type	(Serum, sputum, spinal fluid, etc.)
Sample matrix	(Add rows for each matrix to be validated)

7.4 **Origin of Clinical Samples:**

Sample Material	Source	Description/ Characterization
[insert sample material here]	(Internal or supplier name)	(CDC or ATCC Strain #), prior testing (comparator method or “gold standard”), clinical case definitions, etc.

7.5 Human subjects’ determination # for use of left-over clinical specimens, as applicable:

7.6 **Contrived “spike-in” specimens:**

Contrived “spike-in” specimens	Description
Total positive samples	#

Contrived “spike-in” specimens	Description
Total negative samples	#
Sample volume (units)	#
Sample source/type	(Serum, sputum, spinal fluid, etc.)
Sample matrix	(Add rows for each matrix to be validated)

7.7 **Origin of Contrived “Spike-in” Samples:**

Sample Matrix	Source	Description of “spike in” material	Description of Titrations
[insert Sample Matrix information here]	(Internal or supplier name)	(CDC or ATCC Strain #)	*[insert description of titrations here]*

9.0 **Performance Characteristics:**

NOTE: These definitions are specific for the purpose of this document and are broad to fulfill multiple requirements, including CLIA, for qualitative infectious disease assays. Each laboratory may need to modify calculations found in this table to meet their needs and applications.

TP- True Positive, TN- True Negative, FP- False Positive, FN- False Negative, CV-Coefficient of Variation

Characteristic	Number of samples	Planned Calculations/ Evaluations
Accuracy	(# of positive samples and # of negative samples) will be measured. *NOTE:* Clinical samples or contrived samples in physiological range and near cutoff values may be used.	
Precision/ Reproducibility (Qualitative)	(# of positive samples and # of negative samples) will be measured in (#) separate runs over at least (# 3) days, at least # days apart, by # different operators.	
Precision/ Reproducibility (Raw Value) (e.g., ANI score, CT value)	As above, but evaluating raw, quantitative values, if available, prior to qualitative cut-off for resulting.	
Clinical Sensitivity/ Diagnostic Sensitivity	(# of positive clinical samples) will be measured.	

Characteristic	**Number of samples**	**Planned Calculations/ Evaluations**
Analytic Sensitivity (Qualitative)	(# of samples of different serotypes/genotypes/geographical variants/species expected to be detected) will be measured If not doing a formal LOD study (see in LOD row below) need: (# of quantifiable samples at specified concentration levels near the expected detection limit) will be measured.	
Clinical Specificity/ Diagnostic Specificity	(# of negative clinical samples) will be measured.	
Analytic Specificity	(# of negative samples known to contain potentially cross-reactive agents or analytes) will be measured.	With description of cross-reactive agents or analytes
Limit of Detection (LOD)	Depth of coverage range of (# to #) will be used to determine the bio informatics LOD. Target biological material present in the range (# to #) will be used to determine the biological LOD (e.g., testing/sequencing of serial dilutions).	Bioinformatics LOD = (minimum depth of coverage and consensus % necessary to achieve accuracy) Biological LOD = (minimum amount of target biological material necessary to achieve accuracy)
Evaluation of Assay Cut-off Values (diagnostic assay)	Target biological material present in the range (# to #) will be used to determine the performance of the cutoff value (e.g., CT value).	Method for evaluating assay cutoff value, including equivocal ranges
Reference/Normal value	Determine: Literature review or to generate in-house data In-house: (# of normal population samples) will be evaluated.	Summary of results List of references
Applicable genome region	(#) samples will be sequenced.	Region of the genome in which sequence of acceptable quality was derived in the tested samples.

Characteristic	Number of samples	Planned Calculations/ Evaluations
Clinical validity or Post-test Probability/ Predictive Values	Determine: Literature review describing the assay's clinical validity or calculate predictive values from in-house data based on possible prevalence scenarios (consult with epidemiologists), and link to intended use of assay (section 3.4 above). *NOTE:* If the validation has low power (e.g., n= 5 positive samples tested with 100% sensitivity, 20 negatives with 100% specificity), this analysis should be delayed until there is adequate sample size to be informative.	Positive Predictive Value: TP / (TP + FP), and Negative Predictive Value: TN / (TN + FN) by sample type and/or population or List of references describing clinical validity of the assay

10.0 **Interfering Substances:** If not assessed, state that it is "not assessed" and provide justification why.

8.0

9.0

9.1 If interference is observed during these studies, the interferent should be tested by serial dilutions to determine the lowest concentration that provides interference. Assay limitations may be added to the validation summary.

9.2 Negative interferences:

Potential interfering substance to generate false negative result	Concentration	Volume of Positive Clinical Specimen Diluted, or titered "spike in" specimen	Results
List here in multiple rows: e.g., Hemoglobin, human anti-mouse antibody, rheumatoid factor	High end of clinical range	*[insert volume of positive or spike-in specimen here]*	# detected / total # replicates

9.3 Positive interferences:

Potential interfering substance to generate false positive result	Concentration	Results

Potential interfering substance to generate false positive result	Concentration	Results
List here in multiple rows: e.g., hemoglobin, human anti-mouse antibody, rheumatoid factor	High end of clinical range	# detected / total # replicates

11.0 **Matrix Equivalency (if applicable):**

Characteristic	Number of samples	Planned Calculations/ Evaluations
Other Matrix Equivalency	Measure # of titrations of analyte (high, medium, low, and equivocal if applicable) in # replicates of negative reference matrix in parallel to # replicates of negative test matrix.	

12.0 **Multiplex Assay Performance (if applicable):**

10.0

11.0

11.1 For assays that detect multiple targets, it is necessary to show that high concentrations of one target do not interfere with the detection of other targets.

Characteristic	Number of samples	Planned Calculations/ Evaluations
Multiplex Analytic Sensitivity/Lower limit of Detection	List common co-infections (# of quantifiable samples at specified concentration levels near the expected detection limit) will be measured in presence of high concentrations of the other multiplex target(s).	Describe the analytical sensitivity in the presence of co-infections

13.0 **Plan for Discrepancy Testing (if applicable):**

12.0

12.1 Detail any “tie-breaker” testing to be performed by using additional method, as applicable, for specimens with results discrepant from expected values identified during the validation study. The results from repeated testing, unless repeated testing is included in the test procedures under validation, cannot be included in performance characteristics calculations.

NOTE: This is more applicable for historically characterized isolates than for primary specimens.

14.0 **Handling Outlier Data:**
13.0

13.1 Outlier datapoints are not to be removed from performance calculations, with exceptions that may include major pre-analytical errors (e.g., wrong sample tested). Provide the plan for evaluation of any data points that may be considered "outliers," documented review by the Technical Supervisor prior to the removal from validation calculations, and calculation results provided in the summary report before and after outlier removal.

NOTE: True positive and true negative validation samples must be carefully chosen.

NOTE: All raw data should be included with the Method Validation Report for transparency.

15.0 **Roles and Responsibilities:**
14.0

14.1 (Insert name) is responsible for preparing the Method Validation Plan.
14.2 (Insert name) is responsible for performing the Method Validation.
14.3 (Insert name) is responsible for Document Management.
14.4 (Insert name) is responsible for review and approval of the Validation Protocol prior to testing.
14.5 (Insert name) is responsible for review and approval of the Validation Protocol upon completion.

16.0 **Proposed Timeline:**
15.0

15.1 Identify expected timeframe for each experiment and establish a timeline for completion of the validation and approval of the method for implementation by the laboratory to report results.

15.2 (Insert timeframe) is the expected/allotted amount of time required to perform specimen stability testing in accordance with all established criteria.

17.0 **Instrumentation:**

Name/Model	Serial/ID #	Cal Date	Cal Due Date

Name/Model	Serial/ID #	Cal Date	Cal Due Date
[insert instrument name or model here]	*[insert serial or ID number here]*	*[insert instrument calibration date here]*	*[insert instrument calibration due date here]*

18.0 **Related Documents:**

Document Name	Document Number	Rev. #	Effective date
(Method name) Procedure, if applicable	*[insert document number here]*	*[insert revision number here]*	*[insert effective date here]*
(Comparison Method name) Procedure, if applicable	*[insert document number here]*	*[insert revision number here]*	*[insert effective date here]*

19.0 **Data Management:**

16.0

17.0

18.0

18.1 Describe data management plan, data location, and retention of all validation records (including those used for retrospective analyses).

NOTE: All validation records should be associated with the test procedure and retained for at least 2 years (CLIA) following discontinuation of the test procedure.

20.0 **Bioinformatics Pipeline:**

Name	Version #	Parameter Settings	Developer	Tech Support POC
(List the hardware, software, transmission system, backups, networks)	*[insert version number of bioinformatics pipeline here]*	*[insert information about parameter settings here]*	*[insert name of the developer here]*	*[insert name of tech support POC here]*

21.0 **Training Requirements:**

19.0

20.0

20.1 Personnel performing the Method Validation are required to complete and document training in the test procedure prior to validation. The following staff are trained, and documentation is complete.

Personnel	Type of Training	Date Completed
[insert personnel performing method validation here]	*[insert type of training required here]*	*[insert training completion date here]*

22.0 **Appendices:**
21.0

21.1 Appendix A — Laboratory-Defined Standard Time Points and Temperatures for Specimen Stability

21.2 Appendix B — Specimen Stability

23.0 **Revision History**

Rev #	DCR #	Change Summary	Date
[insert laboratory-specific revision number here]	*[insert laboratory-specific document control number here]*	*[insert laboratory-specific change summary here]*	*[insert laboratory-specific revision date here]*

24.0 **Approval**

Approved By: Date:

Author

Print Name and Title

Approved By: Date:

Team Lead/Supervisor

Print Name and Title

Approved By: Date:

Quality Manager

Print Name

Appendix A - Laboratory-Defined Time Points and Temperatures for Specimen Stability

Molecular assays (all primary specimen types):
NOTE: The times below are conservative and based on a survey of package inserts. The laboratory is responsible for defining their specimen stability criteria and temperature ranges.

Temperature Range	Time Evaluated
2-8°C	RNA: 72 hours
	DNA: 7 days
-20°C or lower	2 months
-70°C or lower	1 year

Appendix B - Specimen Stability

1.0 Specimen Stability:

1.1 Specimen stability testing will provide data to support specimen acceptance criteria. This includes acceptable temperature and time ranges for storage, preservation, and shipment of samples prior to receipt, in addition to stability of samples and their derivatives during the testing.

1.2 **"Bench top" Stability:** Specify stability testing needed for specimens or derivatives as handled during laboratory testing (e.g., an extract stored over a long weekend).

1.3 **Freeze and Thaw Stability:** as applicable to mimic intended specimen handling conditions (e.g., three times frozen at and then thawed [3 freeze-thaw cycles]).

1.4 **Specimen Conditions to be validated**: Refer to standardized timepoints in **Appendix B;** setting criteria outside these parameters would require larger sample size (minimum of 20).

NOTE: Additional conditions and timepoints that are outside of the normal acceptance boundaries of the test system may be evaluated by the laboratory to provide additional data to evaluate test performance.

NOTE: These additional data do not directly extend the range of stability without full validation and procedural approval by the CLIA Laboratory Director (e.g., if acceptable specimen temperature range for a test system is 2-8°C, the laboratory may gather additional data from samples at room temperature.)

Specimen Storage Temperature	**Specimen Condition to be validated: Acceptable time post collection & prior to receipt at CDC (Test Directory Content)**	**Specimen Shipping Condition**	**Time points evaluated in stability study**	**Analytic testing time frame**	**Storage condition during testing (specimens and derivatives)**
Room temperature (15-25°C)	*3 days (serum)*	*Frozen on dry ice*	*3 days, 7 days*	*2 weeks*	*n/a*
Refrigerated (2-8°C)	*7 days (serum)*	*Frozen on dry ice*	*7 days, 14 days*	*2 weeks*	*Serum- 1 day (thawed)*
Frozen (<-20°C)	*2 months (serum)*	*Frozen on dry ice*	*2 months, 6 months*	*2 weeks*	*Serum- 1 month DNA extract – 1 month*
Ultra-low (<-70°C)	*1 year (serum)*	*Frozen on dry ice*	*>1 year*	*2 weeks*	*Serum - indefinitely DNA extract - indefinitely*

1.5 **Specimen Stability Sample Size and Calculations:**

Specimen Condition Evaluated	**Number of samples, temperatures, and time points**	**Planned Calculations**	**Minimum acceptable values**
Plasma, Refrigerated 3 days (List each condition in a separate row)	e.g., 5 positive spike-in samples and 1 negative matrix: Refrigerated 2-8 At: t=0, t=3, Extended timepoints: Room temperature 3 days -Refrigerated 7 days	-AND- Quantitative analysis of unreported raw values (e.g., Ct value): Acceptable mean change from baseline (t=0).	Qualitative: 100% concordance with timepoint zero (small sample size) 95% concordance with sample size 20 or more -AND- Quantitative analysis of unreported raw values (e.g., Ct value): acceptable mean change from baseline. (e.g., within <3.3 Ct) Extended timepoint: descriptive evaluation of challenged samples
Freeze and thaw stability (as applicable to mimic intended specimen handling conditions)	*[insert laboratory specific examples or information here]*	Three times frozen at -80°C and then thawed (3 freeze-thaw cycles) -AND- Quantitative analysis of unreported raw values (e.g., Ct value)	Qualitative: 100% concordance -AND- acceptable mean change from baseline. (e.g., within <3.3 Ct)

Specimen Condition Evaluated	**Number of samples, temperatures, and time points**	**Planned Calculations**	**Minimum acceptable values**
"Bench top" stability- e.g., Extracts, 5 days refrigerated (2-8 C)	e.g., 5 positive spike-in samples and 1 negative matrix: Room Temperature at 15-25 At: t=0, t=3, t=5 Extended timepoints: Room temperature 3 days -Refrigerated 7 days	-AND- Quantitative analysis of unreported raw values (e.g., Ct value): Acceptable mean change from baseline (t=0)	Qualitative: 100% concordance with timepoint zero (small sample size) 95% concordance with sample size 20 or more -AND- Quantitative analysis of unreported raw values (e.g., Ct value): acceptable mean change from baseline. (e.g., within <3.3 Ct) Extended timepoint: descriptive evaluation of challenged samples